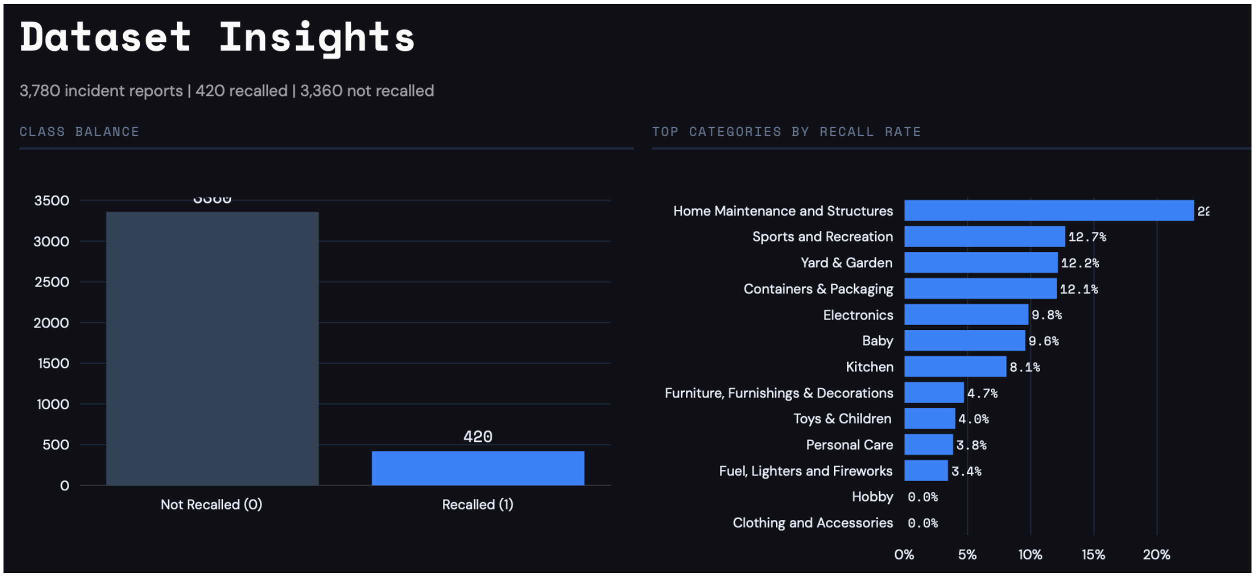
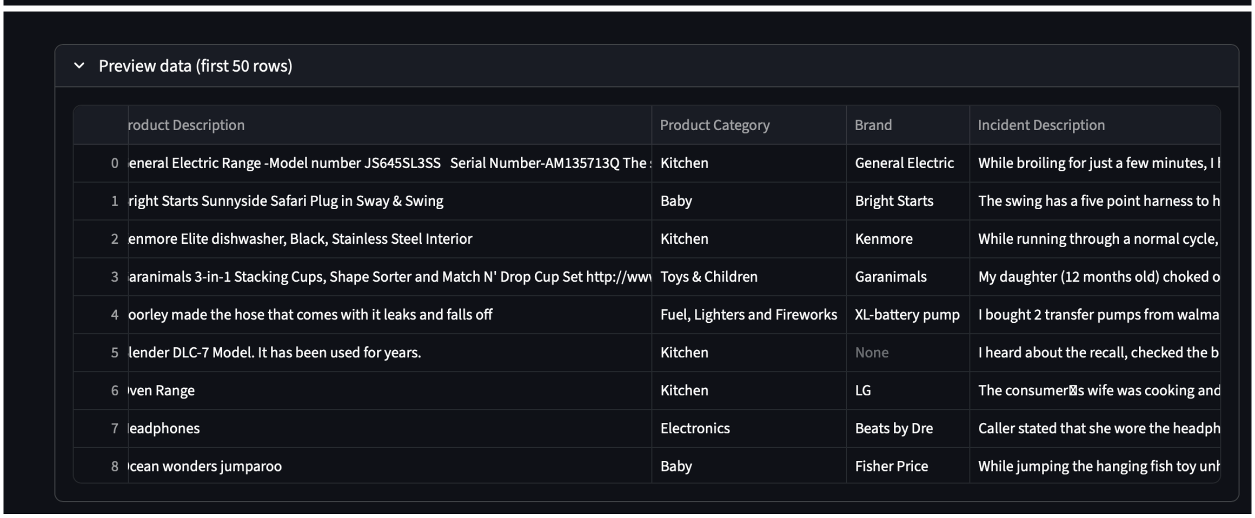
Dataset Insights





Preview data (first 50 rows)

	roduct Description	Product Category	Brand	Incident Description
0	eneral Electric Range -Model number JS645SL3SS Serial Number-AM135713Q The	Kitchen	General Electric	While broiling for just a few minutes, I l
1	right Starts Sunnyside Safari Plug in Sway & Swing	Baby	Bright Starts	The swing has a five point harness to h
2	enmore Elite dishwasher, Black, Stainless Steel Interior	Kitchen	Kenmore	While running through a normal cycle,
3	aranimals 3-in-1 Stacking Cups, Shape Sorter and Match N' Drop Cup Set http://ww	Toys & Children	Garanimals	My daughter (12 months old) choked o
4	oorley made the hose that comes with it leaks and falls off	Fuel, Lighters and Fireworks	XL-battery pump	I bought 2 transfer pumps from walma
5	lender DLC-7 Model. It has been used for years.	Kitchen	None	I heard about the recall, checked the b
6	ven Range	Kitchen	LG	The consumer's wife was cooking and
7	eadphones	Electronics	Beats by Dre	Caller stated that she wore the headph
8	cean wonders jumparoo	Baby	Fisher Price	While jumping the hanging fish toy unl

Category Insights


Category Insights
Explore recall patterns and key risk terms within each product category.
Select a category
Home Maintenance and Structures
TOTAL ROWS
789
RECALL RATE
22.9%
MODEL F1
0.800
MODEL RECALL
0.848
KEYWORDS — HIGHER RECALL RISK
dehumidifier (5.38)
samsung (3.24)
machine (1.66)
cycle (1.45)
pint (1.41)
soleusair (1.36)
deh (1.36)
date (1.33)
spin (1.30)
dehumidifier model (1.27)
45 (1.27)
washer (1.24)
working (1.22)
load (1.20)
spin cycle (1.17)
KEYWORDS — LOWER RECALL RISK
dryer (-1.44)
door (-1.21)
gas (-1.13)
mold (-0.98)
glass (-0.96)
water heater (-0.95)
shower (-0.92)
ladder (-0.82)
cleaner (-0.81)
vacuum (-0.80)
lg (-0.80)
redacted (-0.74)
panel (-0.68)
ge (-0.66)
close (-0.66)
dehumidifier
samsung
machine
cycle
pint
soleusair
deh
date
spin
dehumidifier model
45
washer
working
load
spin cycle
0
1
2
3
4
5
dryer
door
gas
mold
glass
water heater
shower
ladder
cleaner
vacuum
lg
redacted
panel
ge
close
-1.5
-1
-0.5
0
How to read the keyword coefficients
Coefficients come from a Logistic Regression model trained only on text within this category. Positive values mean the keyword pushes the model toward predicting a recall. Negative values push toward not recalled.
These are not standalone risk scores, they reflect the learned association within the full TF-IDF feature space for this category.

Risk Scorer

Risk Scorer

Enter product and incident details to estimate recall risk. Combines the ML model prediction with a hazard-severity analysis for a practical composite score.

Browse by category

Select a product

Pick a category first

Load Selected Product | Load Random Example | Clear Fields

Product Description

KIDDE dual sensor smoke/fire detector/alarm Model #PI 2010

Incident / Complaint Text

In 2015, I had a professional electrician install 2 new hardwired smoke/fire detectors.
They are: KIDDE Dual Sensor Smoke & Fire hardwired detectors Model

Brand

KIDDE

Manufacturer / Importer / Private Labeler

UTC Fire & Security

Model Name or Number

PI2010

Product Category

Home Maintenance and Structures

Product Sub Category

Home Security

Product Type

Fire or Smoke Alarms (702)

Estimate Recall Risk

70%

Composite Risk Score · **High**

ML MODEL PROBABILITY

92.0%

HAZARD SEVERITY SIGNAL

40%

Suggested Action: Escalate for detailed safety review. Gather related incident data and consider contacting the manufacturer.

› What influenced this score?

› Why two signals?

What influenced this score?

Composite Score Breakdown

- ML model contribution: 92.0% x 58% weight = 53.4%
- Severity contribution: 40% x 42% weight = 16.8%

Hazard Language Detected

- High-severity terms (2): burn, fire, kid, smoke
- Moderate terms (2): defect, recall, recalled

Other Factors

- Category *Home Maintenance and Structures* has a base recall rate of 22.9%
- Model number provided; may match known recalled products

Why two signals?

The ML model was trained to predict **formal CPSC recalls**, not whether a product is dangerous. Most dangerous incidents never lead to a formal recall because recall decisions depend on factors the model cannot see (legal negotiations, complaint volume, manufacturer cooperation).

This means a genuinely alarming incident (like a battery explosion causing injury) can get a low ML probability if products in that category were rarely recalled in the training data.

To make the tool practically useful, we combine two signals:

- **ML Model Probability** : learned patterns from historical recall data (brand, model, category, text features)
- **Hazard Severity Signal** : rule-based analysis of how dangerous the incident language sounds (critical terms like "death", "explosion", "hospitalized" score higher than moderate terms like "defective" or "pain")

The composite score gives more weight to the severity signal when the incident language is alarming.

Model Details

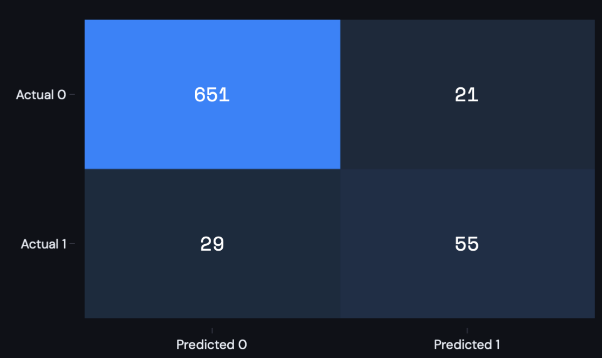
RecallGuard

A machine-learning system that estimates product recall risk from CPSC incident reports. Built with TF-IDF feature engineering, multi-model comparison, SMOTE oversampling, and threshold-tuned classification.

BEST MODEL PERFORMANCE

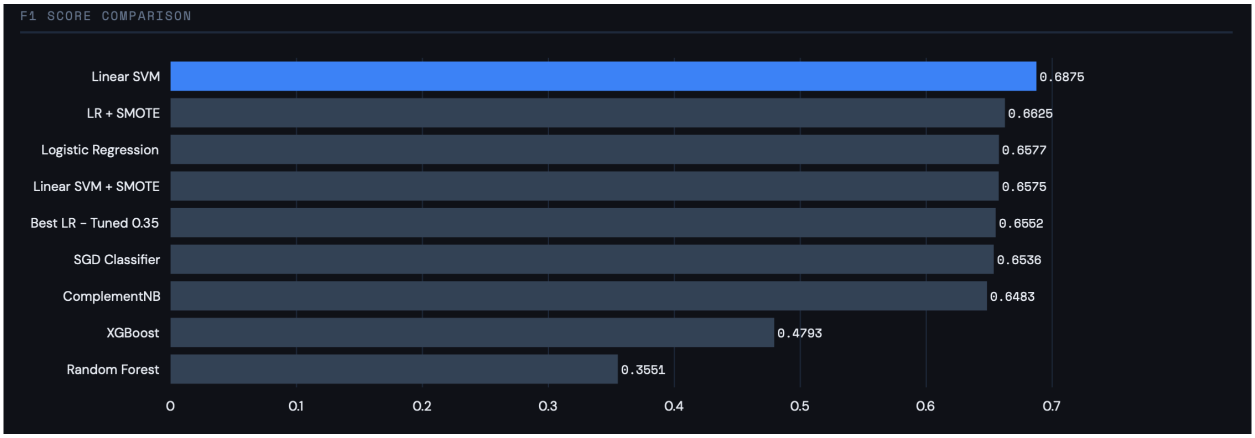
BEST MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
Linear SVM	0.9339	0.7237	0.6548	0.6875

CONFUSION MATRIX



FULL MODEL COMPARISON

Model	accuracy	precision	recall	f1
Linear SVM	**0.9339**	0.7237	0.6548	**0.6875**
LR + SMOTE	0.9286	0.6974	0.6310	0.6625
Logistic Regression	0.9325	0.7538	0.5833	0.6577
Linear SVM + SMOTE	**0.9339**	0.7742	0.5714	0.6575
Best LR - Tuned 0.35	0.9206	0.6333	**0.6786**	0.6552
SGD Classifier	0.9299	0.7246	0.5952	0.6536
ComplementNB	0.9325	0.7705	0.5595	0.6483
XGBoost	0.9167	0.7838	0.3452	0.4793
Random Forest	0.9087	**0.8261**	0.2262	0.3551



How to interpret these metrics

Accuracy : Fraction of all predictions that were correct. High here because most products are *not* recalled (class imbalance).

Precision : Of the products the model flagged as recall-likely, how many actually were recalled. Higher precision = fewer false alarms.

Recall (Sensitivity) : Of products that *were* recalled, how many did the model correctly catch. Higher recall = fewer missed recalls.

F1 Score : Mean of precision and recall. The primary metric for this project because both false alarms and missed recalls matter.

Confusion Matrix : Top-left is True Negatives, bottom-right is True Positives. Off-diagonal cells are errors.

SMOTE Models : Synthetic Minority Over-sampling Technique generates synthetic examples of the minority class (recalled) to address class imbalance.

Threshold Tuning : Instead of using the default 0.50 cutoff for Logistic Regression, the tuned version picks the threshold that maximizes F1 on out-of-fold cross-validation predictions.

XGBoost : Gradient-boosted decision tree ensemble with scale_pos_weight set to the class imbalance ratio.

Random Forest : Bagged decision tree ensemble with balanced class weights.

SGD Classifier : Stochastic Gradient Descent with log loss (equivalent to logistic regression optimized via SGD), using balanced class weights.